

THREAD CLASS METHOD AND CONSTRUCTORS:

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Constructors :

Thread Class: Methods and Constructors in Java

The **Thread** class in Java is used to create and manage threads. It belongs to the **java.lang** package and provides various **constructors** and **methods** to control thread execution.

1. Thread Class Constructors

The Thread class provides multiple constructors to create threads in different ways:

(i) Thread()

Creates a new thread without a name and without a Runnable object.

(ii) Thread(String name)

Creates a new thread with a specified name.

(iii) Thread(Runnable target)

Creates a thread that will execute the run() method of the provided Runnable object.

(iv) Thread(Runnable target, String name)

Creates a thread with a specific Runnable target and assigns a name.

METHODS :

2. Important Methods of Thread Class

The Thread class provides several methods to control and manage threads.

(i) start()

Starts a new thread and calls the run() method.

```
t1.start();
```

(ii) run()

Contains the code that runs when the thread is started.

```
public void run() {  
    System.out.println("Thread is running...");  
}
```

(iii) sleep(long milliseconds)

Pauses the thread execution for the specified time.

Example:

```
try {  
    Thread.sleep(2000); // Sleeps for 2 seconds  
} catch (InterruptedException e) {  
    e.printStackTrace();  
}
```

(iv) join()

Waits for a thread to finish before proceeding.

Example:

```
Thread t1 = new Thread(() -> {  
    try {  
        Thread.sleep(3000);  
    } catch (InterruptedException e) {  
        e.printStackTrace();  
    }  
    System.out.println("Thread finished.");  
});  
t1.start();  
try {  
    t1.join(); // Waits for t1 to finish  
} catch (InterruptedException e) {  
    e.printStackTrace();  
}  
System.out.println("Main thread continues...");
```

(v) yield()

Pauses the current thread to allow other threads of the same priority to execute.

```
Thread.yield();
```

(vi) setName(String name) and getName()

Sets and gets the thread name.

```
t1.setName("CustomThread");
System.out.println(t1.getName()); // Output: CustomThread
```

(vii) setPriority(int priority) and getPriority()

Sets and gets thread priority (1 to 10, where **5 is default**).

```
t1.setPriority(Thread.MAX_PRIORITY);
System.out.println(t1.getPriority()); // Output: 10
```

(viii) isAlive()

Checks if the thread is still running.

```
if (t1.isAlive()) {
    System.out.println("Thread is still running...");
}
```

(ix) isDaemon() and setDaemon(boolean on)

Marks a thread as a daemon (background service thread).

Example:

```
t1.setDaemon(true);
System.out.println(t1.isDaemon()); // Output: true
```

(x) interrupt() and isInterrupted()

Interrupts a sleeping or waiting thread.

```
t1.interrupt();
System.out.println(t1.isInterrupted()); // Output: true
```

➤ Inter Thread Communication:

1. Wait()
2. Notify()
3. NotifyAll()